Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

This indicates that R2 is only forming an adjacency with R1 because only the interface connected to R1 (S0/0/0) has had the passive status removed. Because R2's interface to R3 (S0/0/1) is still passive, they cannot exchange Hello packets or form a neighbour relationship

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1
R2 (config-router)# no passive-interface s0/0/1

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The standard setting of 100Mb/s treats all higher-speed links as having the same cost of 1, preventing OSPF from accurately calculating the most efficient path.

2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24 : R1 S0/0/1 + R3 G0/0 (781+1 = 782)

192.168.23.0/30 : R1 S0/0/1 and R3 S0/0/1 (781+64 = 845)

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

781+781 = 1562 . Since both serial link now has a cost of 781, and the route to the network 192.168.23.0/24 travels over two serial links.

2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost = 781 + 781 + 1 = 1563, which is smaller than 1565+1 = 1566

R1-S0/0/0 R2-S0/0/1 R3-G0/0 R1-S0/0/1 R3-G0/0

2+

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   It controls the designated router and backup designated router election process on a multi-access network. If the router ID is associated with an active interface, it can change if the interface go down. As a result, it should be using IP address of a loopback interface.

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2. Why is the DR/BDR election process not a concern in this lab?

   The process is only an issue on a multi-access network such as Ethernet or Frame Relay. Point-to-point links are used for serial links, therefore no DB/BDR election is performed.

   4

3. Why would you want to set an OSPF interface to passive?

   It would eleminate unnecessary OSPF routing information on that interface, freeing up bandwidth.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   The use of areas makes OSPF hierarchical, which allow for route summarization and limiting the scope of routing updates

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be used.

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